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Task 03: Scenario-Based Task: Sales Analysis & Prediction for Supermarket

Assume the role of a Data Analyst working for a supermarket chain. The management has provided you with historical sales data to understand sales patterns and support data-driven business decisions.

Business Scenario:

The supermarket operates across multiple branches and cities and offers various product categories with different pricing and discount strategies. Management wants to:

- Understand what factors influence total sales
- Predict future sales using data
- Identify areas where revenue and profitability can be improved

Your Objectives:

1. Analyse the dataset to understand sales behaviour
2. Build a Linear Regression model to predict Total Sales
3. Evaluate model performance using appropriate error metrics
4. Provide actionable business suggestions based on your analysis

Tasks to Perform:

1. **Data Understanding & Cleaning**
    - Describe the dataset and variables
    - Handle missing or inconsistent values (if any)
2. **Exploratory Data Analysis (EDA)**
    - Identify key factors affecting sales
    - Analyze relationships between price, quantity, discount, and sales

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3. **Model Development**
    - Define independent (X) and dependent (y) variables
    - Perform train–test split (70–30 or 80–20)
    - Implement:
        - Simple Linear Regression
        - Multiple Linear Regression
4. **Model Evaluation**
    - Calculate and explain:
        - Mean Absolute Error (MAE)
        - Mean Squared Error (MSE)
        - Root Mean Squared Error (RMSE)
        - R^2 Score
5. **Interpretation & Insights**
    - Interpret regression coefficients
    - Identify which variables most impact sales
    - Discuss overfitting vs underfitting (I'll add one more document on this concept)

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Dataset Overview (What's Inside)

Columns Available (Regression-Ready)

- Invoice_ID
- Branch
- City
- Customer_Type
- Gender
- Product_Category
- Unit_Price
- Quantity
- Discount
- Discount_Amount
- Net_Sales
- Tax_7.25_percent
- **Total_Sales** *(Target variable)*
- Rating

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